A Two-Stage, Leakage-Aware Framework for Early Academic Risk Detection in Undergraduate Engineering Cohorts

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***Abstract*—Timely identification of students at risk of meaningful academic decline enables targeted advising and reduces avoidable failures. We present a leakage-aware, two-stage machine learning framework that first forecasts subject-wise course marks using only information available up to the end of prior semesters, and then translates these forecasts into cohort-relative risk signals suitable for operational interventions. Using anonymized records for 905 undergraduate engineering students with 56 features (demographics, attendance and past marks), we model four core subjects independently and compute predicted cohort percentiles. A student is labeled "at risk" if the predicted cohort percentile drops by 10 or more points relative to the prior semester; Semester 3 is used herein as an illustrative case study to demonstrate the approach. Across subjects, a Voting Regressor (Ridge + Lasso + ElasticNet) with One-Hot encoding and Robust scaling yields test MAEs between 5.72–7.10. A stacking classifier (CatBoost, EasyEnsemble, Balanced-Bagging LGBM with a logistic meta-learner) attains a test accuracy of 0.674, recall of 0.657, and an F1-score of 0.438 at a threshold chosen to prioritize recall. We discuss leakage prevention, deployment, ethical considerations, and directions for multi-institution validation. A lightweight web implementation of the pipeline is accessible online.**

***Index Terms*—student performance prediction, early warning systems, ensemble learning, stacked classification, leakage prevention**

I. Introduction

Predicting student performance is a longstanding challenge in educational data mining and learning analytics. Accurate forecasts of academic outcomes can provide institutions with early warning signals and enable targeted interventions before students experience significant academic decline. In particular, engineering programs present a high cognitive load, where early detection of performance risks is essential to reduce dropout rates and improve long-term success.

This work addresses the problem through a two-stage machine learning framework. First, we build subject-wise regression models to predict core Semester 3 marks using prior performance in Semesters 1 and 2. Second, we design a classification model that detects students at risk of a significant performance drop. In this study, *risk* is formally defined as a decline of at least ten percentile points between a student's actual Semester 3 percentile and their Semester 2 percentile. This formulation moves beyond raw score prediction by quantifying academic decline in relative terms. By jointly evaluating both regression and classification tasks, the system aims not only to forecast marks but also to assess the likelihood of adverse academic trajectories.

An interactive demonstration of the system is publicly available [1].

II. Related Work

Research on predicting student outcomes and building early-warning systems spans two closely linked communities: Educational Data Mining (EDM) and Learning Analytics (LA). Comprehensive surveys summarize decades of work on predicting grades, retention, and dropout, and highlight common features (prior grades, attendance, learning management system logs) and model families (tree-based ensembles, support vector machines, neural networks) [2], [3].

A substantial body of applied work focuses on early-warning systems that provide actionable signals for advisors. These studies evaluate institutional systems that combine predictive models with workflow support, and emphasize timeliness, interpretability, and the trade-off between precision and recall in intervention settings [4], [5].

In terms of modeling approaches, ensemble techniques such as boosting and stacking are widely used in student-performance prediction tasks. Recent studies show that combinations of decision-tree ensembles with gradient boosting frameworks such as XGBoost [6], LightGBM, or CatBoost [7] often improve robustness and predictive accuracy over single learners, particularly on heterogeneous educational datasets [3].

Model interpretability and explainability are increasingly treated as essential components of deployed educational systems. SHAP (SHapley Additive exPlanations) has become a widely adopted tool for producing per-student explanations and supporting advisor trust and accountability [8].

Finally, data leakage and related preprocessing errors are well known to inflate reported performance in predictive modeling. The literature stresses leakage-aware experimental design (temporal splits, feature engineering restricted to pre-prediction information, and careful validation) as essential for reliable evaluation and real-world deployment [9].

III. Methodology

A. Dataset

The experiments use an anonymized dataset of 905 undergraduate engineering students across three semesters. Data, sourced from university records and institutionally cross-checked, is highly reliable. Full documentation and an anonymized copy are in the project repository [1]. The dataset has 56 columns spanning demographics, administrative metadata, and course-level academic records (theory/practical marks, attendance where available). Marks and attendance are on a 0–100 scale. Attendance was not recorded for some workshops or for Semester 3 (by design); recorded fields contain no nulls.

Privacy: enrollment numbers and names were replaced with synthetic Student IDs for grouping. Derived demographics (gender, religion) were algorithmically inferred and treated as noisy variables, used only for exploratory/fairness analyses.

The binary risk label used for classification (Risk Flag) is defined as a drop of at least 10 percentile points between the student's actual Semester 3 percentile and their Semester 2 percentile. At training and evaluation time, the label is computed from observed Semester 3 marks; at deployment time, Semester 3 marks are not yet available, and the trained classifier instead consumes predicted Semester 3 marks (from Stage 1) as features to produce a risk estimate. The positive class ("at risk") comprises approximately 20% of the dataset, creating an imbalanced classification task (roughly 20% positive, 80% negative).

B. Preprocessing and Feature Engineering

Preprocessing and feature decisions were applied model-specifically and are summarized below.

a) Exclusions (with rationale): To maintain transparency and avoid confounding signals, the following were excluded:

- Student names and original enrollment identifiers (privacy).
- Semester 3 marks: these are the prediction targets and were therefore never used as inputs.
- Administrative groupings (divisions, roll numbers; e.g., Div-2, Div-3, Roll-2, Roll-3): encode cohorting and duplicate performance signals, thus excluded for relevance/redundancy.
- Workshop-only practicals and checkbox subjects: not informative for core targets; retained in repository for exploratory use but excluded from primary pipelines.

b) Correlation/multicollinearity: Redundancy among predictors was checked via pairwise correlation and variance inflation factors (VIF), separately for Stage 1 (subject-wise regression inputs) and Stage 2 (classification inputs built from regression outputs + prior predictors).

Figure 1 shows correlation structure for Stage 1 predictors, highlighting strong redundancy among subject-wise marks and attendance features. Figure 2 shows the corresponding diagnostics for Stage 2 inputs, which include predicted marks and aggregated features.

As in standard practice, some features exceeded correlation/VIF thresholds, indicating possible multicollinearity. However, in both stages, their removal consistently reduced predictive accuracy (higher MAE for regression and lower recall for classification). We therefore retained all features, documenting this choice for reproducibility.

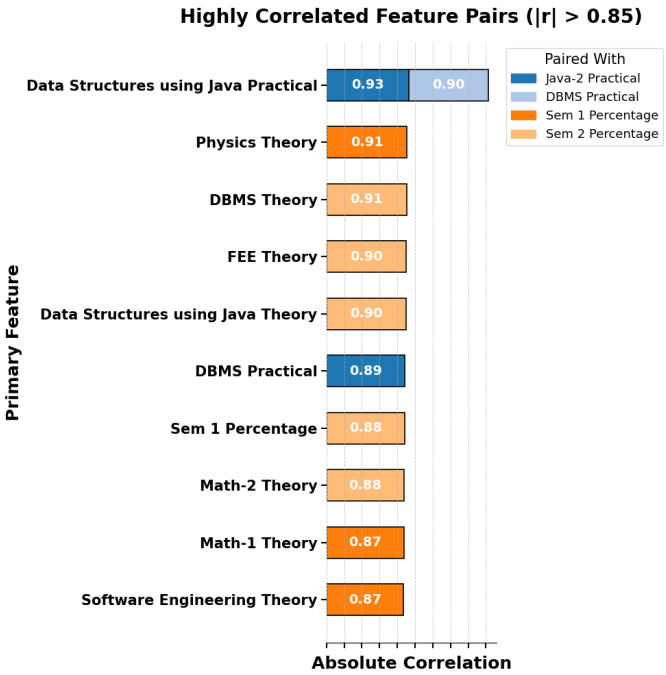

Fig. 1: Stage 1: highly correlated feature pairs ($|r| > 0.85$) among regression predictors.

c) Transformations and derived features: **Stage 1 – Subject-wise Regression**

- Continuous predictors (marks, attendance, derived aggregates) were on a 0–100 scale. For linear base learners in the final Voting Regressor (Ridge, Lasso, ElasticNet), which are sensitive to feature magnitude, numerical features were transformed using RobustScaler. Tree-based learners evaluated during model selection are scale-invariant and did not require this transformation.
- Categorical features (branch, mentor) were one-hot encoded for models that require explicit numeric inputs (for example, Ridge, Lasso, logistic regression). Demographic variables (gender, religion) were algorithmically inferred and excluded from all model pipelines following exploratory analysis, which found them uninformative. Tree-based learners and CatBoost received categorical inputs directly or used their native categorical handling.
- We derived Semester 1–2 summary features such as mean attendance and prior-semester percentiles. No Semester 3-derived values were used as inputs to the regression models.

Stage 2 – Academic Risk Classification

- Predicted Semester 3 subject marks from Stage 1 (Digital Electronics, Math-3, Full Stack Development, and

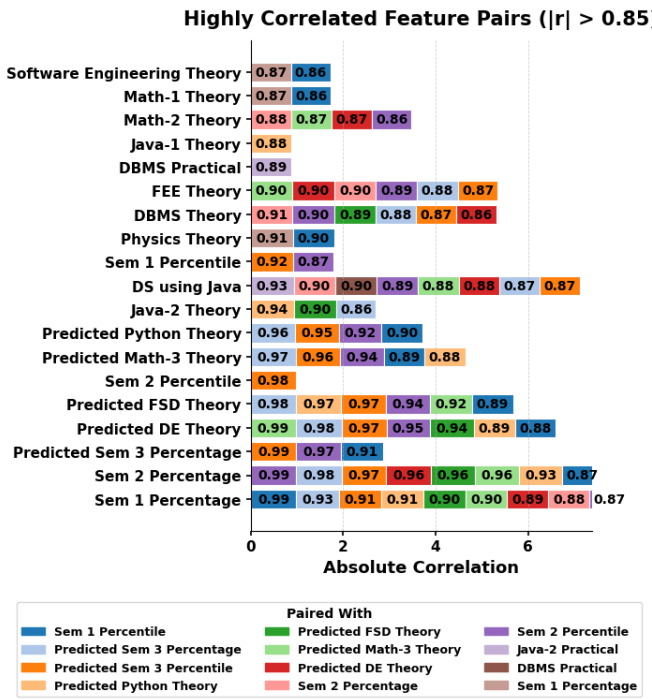

Fig. 2: Stage 2: highly correlated feature pairs for classification inputs (OOF predicted marks + Semester 1/2 aggregates).

Python) were aggregated to form a predicted total score.

- From the predicted total score, we computed a *predicted Semester 3 percentile* within the cohort.
- The actual Semester 2 marks were used to calculate a *Semester 2 percentile* within the cohort.
- The difference between predicted Semester 3 percentile and actual Semester 2 percentile was computed, and a *predicted student risk* feature was created as a Boolean flag: true if the percentile dropped by 10 or more points, and false otherwise.
- Although this derived risk feature was noisy and had inaccuracies, it provided informative signals correlated with student outcomes. Importantly, this input feature is derived from Stage 1 predicted marks, which in turn were trained using only Semester 1 and Semester 2 information (via the out-of-fold protocol described in the following section). No actual Semester 3 outcomes flow into classifier inputs, ensuring leakage-free training.
- This predicted risk feature, along with original Semester 1 and 2 data, was used as input to the Stage 2 classification model without introducing leakage.

C. Experimental Setup

The dataset was split once (fixed random seed) into training (80%) and test (20%) partitions. These partitions were reused across all experiments. The test partition was strictly held out from feature engineering, model selection, and threshold tuning.

Because the pipeline is two-stage (subject-wise regression followed by classification), we used an explicit out-of-fold protocol to avoid optimistic bias when classifier inputs are derived from regressor outputs:

1) On the training partition, regressors for each subject were trained using repeated 5-fold cross-validation. For each fold, the regressor trained on $k-1$ folds was used to predict the held-out fold; these held-out predictions were aggregated into out-of-fold (OOF) predicted-mark features for every training sample.
2) The classifier was trained on the training partition using only the OOF predicted-mark features together with original Sem1/Sem2 predictors. Classifier selection and threshold tuning used cross-validation.
3) For final evaluation, regressors were retrained once on the full training partition (never on test set) and used to generate predicted marks for the held-out test partition; those predictions were then passed to the trained classifier for test-time evaluation.

Regressors were not retrained on test prior to producing classifier features; this rule was enforced throughout to prevent leakage of true Semester 3 outcomes into classifier training.

D. Models and hyperparameter search

For each Semester 3 subject (Math-3, Digital Electronics, Full Stack Development, Python), we evaluated over 50 candidate regression pipelines drawn from the following families: linear models (Ridge, Lasso, ElasticNet, ordinary least squares, quantile regression), support-vector regressors, tree ensembles (RandomForest, ExtraTrees, HistGradientBoosting), and gradient-boosting frameworks (XGBoost, LightGBM, CatBoost). We also evaluated meta-ensembles, including stacking and voting ensembles that combine linear and tree-based learners.

Hyperparameter tuning was applied pragmatically according to model complexity. Full candidate model lists and validation metrics are documented in the project Repo [1].

For risk detection we evaluated over 50 classification pipelines, including class-balancing strategies, ensembles, and stacked architectures. The selected solution was a StackingClassifier with base learners CatBoost, EasyEnsemble [10], and a balanced LightGBM bagging ensemble, and a logistic-regression meta-learner with `predict_proba` passthrough. For the classifier, feature selection was applied in two stages: mutual information filtering to retain the top 40 features, followed by Boruta [11] applied to this subset to identify the final feature set. Decision thresholds were tuned by sweeping values in the range $[0.2, 0.7]$ under 5-fold stratified cross-validation, selecting the threshold that maximized recall, with F1 as a secondary criterion.

Although many candidates were tuned, the final regression and classification pipelines used in the downstream experiments were selected based on the cross-validation performance.

E. Evaluation metrics

Regression performance is reported using Mean Absolute Error (MAE) on cross-validation and on the held-out test

partition. MAE was chosen because it is measured in the same units as student marks and is directly interpretable by institutional stakeholders.

Classification performance is reported using accuracy, precision, recall, and F1-score. Model selection prioritized recall, as identifying at-risk students was critical to ensure timely intervention, and the dataset's class imbalance (approximately 20% positive / 80% negative) further emphasized the need to minimize missed cases.

F. Implementation and reproducibility

All experiments, seeds, and scripts to reproduce the results are available in the repository. Experiments were run with fixed random seeds to ensure repeatability. Computational environment and package versions are also listed in the repository [1].

IV. Results

A. Stage 1: Subject-wise Regression

Four independent regression tasks were defined (Math-3, Digital Electronics, Full Stack Development, and Python).

Pipeline Info:

- **Pipelines evaluated per subject:** 56
- **Best-performing pipeline across all subjects:** A pipeline using the Voting Regressor with the following base models:
  - Ridge Regression
  - Lasso Regression
  - ElasticNet Regression
- **Preprocessing steps included in the pipeline:**
  - OneHotEncoder for categorical features
  - RobustScaler for numerical features
- **Validation strategy:** Repeated 5-Fold cross-validation
- **Hyperparameter optimization:** Bayesian Optimization with BayesSearchCV [12]

Table I reports the test-set MAE for each subject. The pipeline described above consistently achieved the best performance across all four tasks compared to other pipelines evaluated.

TABLE I: Training and test Mean Absolute Error (MAE) for best regression pipelines by subject. Training MAE refers to the cross-validated MAE of the winning pipeline on the training partition.

Subject	Train MAE	Test MAE
Digital Electronics	7.1206	7.1022
Math-3	6.3179	6.9753
Full Stack Development	6.5837	6.6572
Python	5.1607	5.7187

Figures 3, 4, 5, and 6 present subject-specific MAE comparisons by algorithm. Each graph illustrates that this pipeline consistently achieves lower errors across all four subjects compared to other individual learners and ensemble approaches.

Per-subject MAE varies meaningfully, with Python lowest (5.72) and Digital Electronics highest (7.10). The Voting Regressor remained the top performer across all four targets, though its margin over the next-best learner differed by subject.

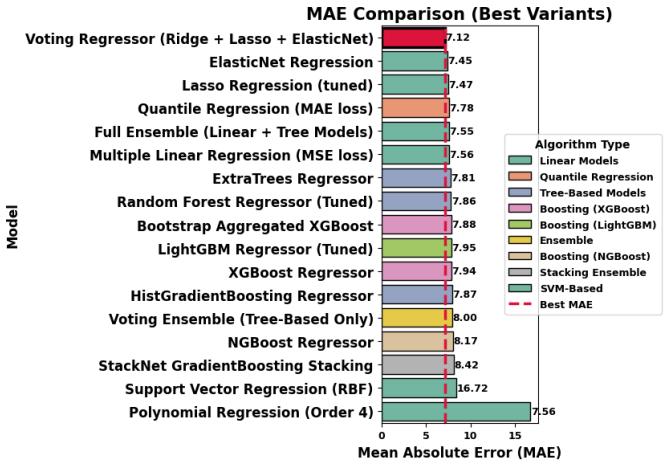

Fig. 3: Digital Electronics: Comparison of MAE across evaluated algorithms.

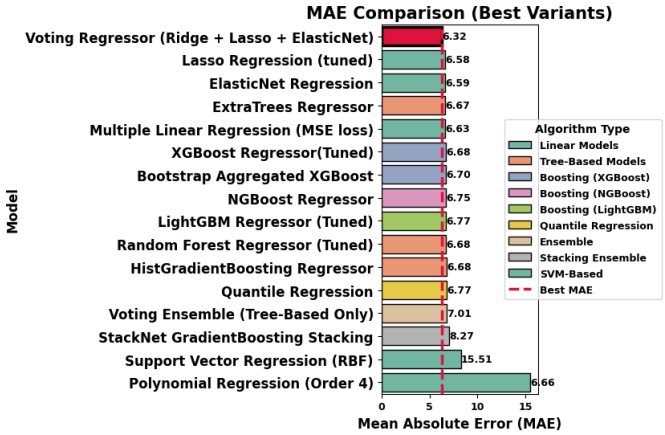

Fig. 4: Math-3: Comparison of MAE across evaluated algorithms.

B. Stage 2: Academic Risk Classification

Predicted Semester 3 marks were aggregated and converted to percentiles. A binary flag was set if the percentile dropped by $\geq$10 points from Semester 2. These features, with original Semester 1 and Semester 2 predictors, formed the Stage 2 classifier inputs.

Model Info:

- **Final model:** Stacking Classifier with base learners CatBoost, EasyEnsemble, and Balanced Bagging (LGBM), and meta-learner Logistic Regression with `predict_proba` passthrough.
- **Preprocessing and feature selection:** OneHotEncoder for categorical features; Mutual Information selection (top 40 features) followed by Boruta applied to this subset for final feature selection.

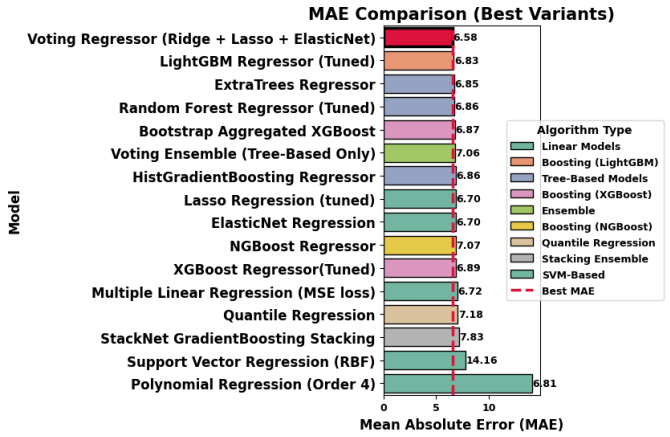

Fig. 5: Full Stack Development: Comparison of MAE across evaluated algorithms.

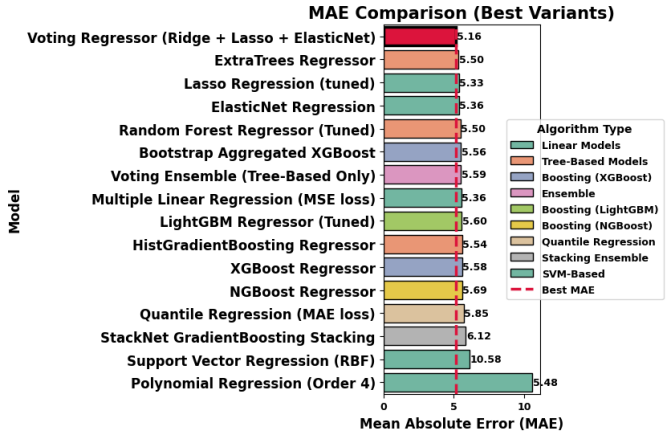

Fig. 6: Python: Comparison of MAE across evaluated algorithms.

- **Cross-validation and threshold tuning:** 5-Fold Stratified Cross-Validation with a threshold sweep in the range $[0.2, 0.7]$, prioritizing recall with F1 as a secondary criterion.

Figure 7 presents a comparison of accuracy, precision, recall, and F1 score across different classification model families. The Stacking Classifier consistently outperformed baseline models such as decision trees, linear classifiers, and gradient boosting methods.

Table II summarizes the evaluation metrics for the final model on both the training and held-out test partitions. The modest gap between training and test metrics (F1 of 0.4793 vs. 0.4381) indicates that the model generalizes without severe overfitting, despite the class imbalance and the recall-prioritized threshold.

C. Model Explainability with SHAP

SHAP (SHapley Additive exPlanations) was used to interpret feature contributions in both regression and classification stages. Across all subject-level regression models (Figures 8, 9, 10, 11), past courses consistently appeared among the top predictors; Fundamentals of Electrical and Electronics (FEE) features consistently ranked highest for Digital Electronics, serving as a representative example. This recurring trend underscores how earlier academic performance strongly shapes later outcomes.

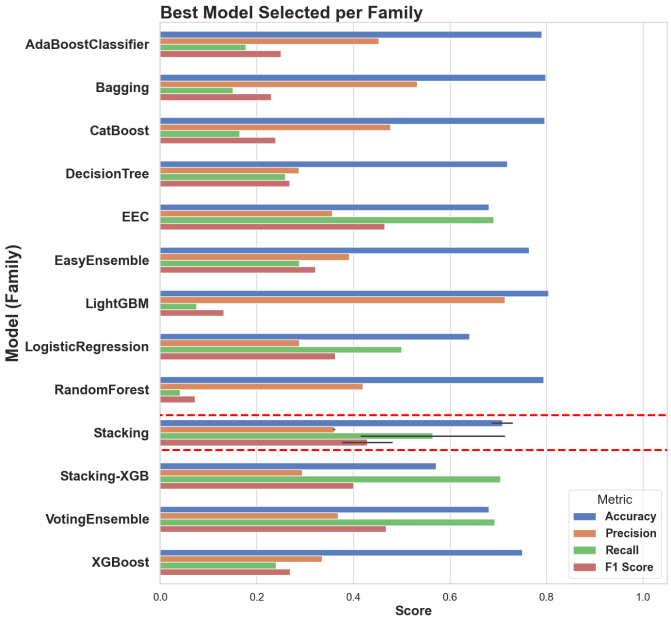

Fig. 7: Comparison of classification performance metrics across evaluated algorithms.

TABLE II: Training and test classification metrics for the final risk detection model.

Metric	Training	Test
Accuracy	0.6878	0.6740
Precision	0.3611	0.3286
Recall	0.7123	0.6571
F1 Score	0.4793	0.4381

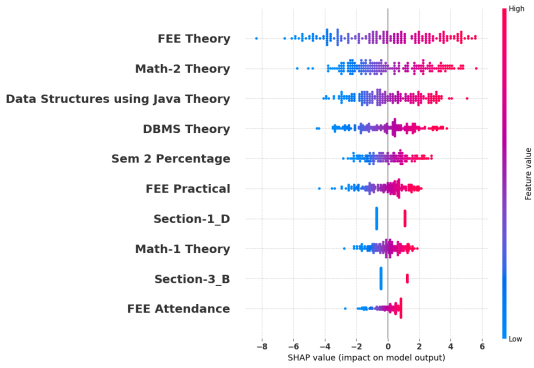

Fig. 8: SHAP for Digital Electronics: Top 10.

For the stage-2 academic risk classifier, Figure 12 shows that SHAP analysis identified Semester 2 theory subject marks as the largest contributors to the predicted risk.

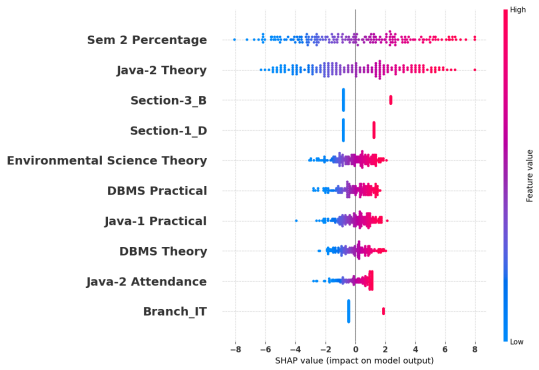

Fig. 9: SHAP for Mathematics 3: Top 10.

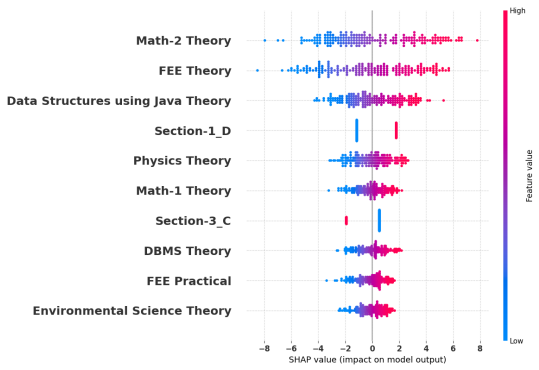

Fig. 10: SHAP for Full Stack Development: Top 10.

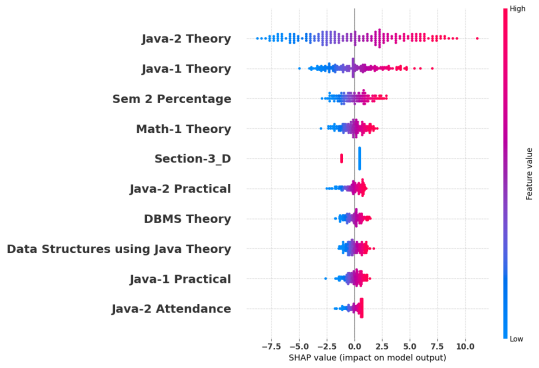

Fig. 11: SHAP for Python: Top 10.

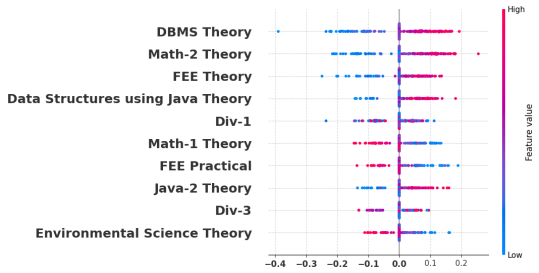

Fig. 12: SHAP for stage-2 risk classification: Top 10.

V. Discussion and Limitations

The regression models achieved test mean absolute errors between 5.72 marks and 7.10 marks, which are within a range meaningful for early academic forecasting in undergraduate cohorts. The classification model prioritized recall (0.657) given the critical need to identify at-risk students early, despite the class imbalance of approximately 20% at risk. This trade-off reflects the operational context in which false negatives (missed at-risk cases) are more harmful than false positives (unnecessary interventions).

The leakage-aware design ensured that future data (Semester 3 marks) did not inform model training, preserving temporal validity. Explainability tools, such as SHAP analysis, support adoption by providing insights into student-level predictions without compromising data privacy.

However, limitations remain. The dataset originates from a single institution and represents one cohort of students enrolled in different, though closely related, computer science–focused branches. While all students share the same core subjects across the first three semesters—which facilitates comparison and model training—the generalizability of the findings to other universities, disciplines, or academic structures remains untested. Additionally, demographic features such as gender and religion were inferred algorithmically, introducing potential noise. Future work should explore the application of this framework across multiple institutions and academic programs, incorporate more diverse and longitudinal datasets, and evaluate the system's effectiveness in real-time operational settings.

Across SHAP analyses for all four regression models and the stage-2 risk classifier, demographic features such as gender and religion did not emerge as important predictors. While these variables showed some noise, their limited influence may be interpreted as suggestive of low demographic bias in the available institutional data [13]. Although some batch identifiers appeared sporadically in SHAP plots, these are not meaningful since student batches are reshuffled by merit each semester.

VI. Conclusion

This paper presented a two-stage, leakage-aware machine learning framework for early academic risk detection in undergraduate engineering cohorts. By combining subject-wise regression forecasts with a stacking classifier that emphasizes recall, the approach provides actionable risk signals while maintaining temporal integrity and explainability. The models demonstrated reliable predictive performance within a controlled dataset, and the framework's structure supports scalable adaptation to other settings.

The work contributes to educational data mining by offering a practical methodology for institutions to identify at-risk students early, enabling timely interventions. Future research will focus on validating the framework across diverse academic contexts, improving feature engineering, and integrating the system into operational advisory workflows to support student success at scale.

ACKNOWLEDGMENT

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